Course Directive
ID607001: Introductory Application Development Concepts
Semester One, 2025

Course Information

Level:	6
Credits:	15
Prerequisite:	ID511001: Programming 2
Timetable:	Rōpū Kōwhai: Monday 1.00 PM D202 and Thursday 1.00 PM D202

Teaching Staff

Name:	Grayson Orr
Position:	Principal Lecturer and Second/Third-Year Coordinator
Office Location:	D309
Email Address	grayson.orr@op.ac.nz

Course Dates

Term 1 (8 weeks):	17 February - 11 April
Term 2 (8 weeks):	28 April - 19 June

Public Holidays and Anniversary Days

A list of public holidays and anniversary days can be found here - https://www.op.ac.nz/students/importantdates

Aims

To introduce the concepts of application development including algorithms, data structures and design patterns that are required to use a simple, industry-relevant development framework.

Learning Outcome

At the successful completion of this course, learners will be able to:

1. Design and build secure applications with dynamic database functionality following an appropriate software development methodology.

Assessments

Assessment	Weighting	Due Date	Learning Outcome
Practical	20%	Thursday of Week 11 at 2.59 PM	1
Project	80%	Monday of Week 15 at 7.59 AM	1

Grade Table - Criterion Referenced

Grade	Mark Range
A+	Met all course requirements-mark in range [90-100]
A	Met all course requirements-mark in range [85-89]
A-	Met all course requirements-mark in range [80-84]
B+	Met all course requirements-mark in range [75-79]
B	Met all course requirements-mark in range [70-74]
B-	Met all course requirements-mark in range [65-69]
C+	Met all course requirements-mark in range [60-64]
C	Met all course requirements-mark in range [55-59]
C-	Met all course requirements-mark in range [50-54]
D	There at end. Did not meet course requirements. Mark in range [40-49]
E	There at end. Did not meet course requirements. Mark in range [0-39]

Provisional Schedule

Week	Topics
1/Tahi	GitHub and JavaScript
2/Rua	REST, Express, HTTP Methods, HTTP Status Codes and HTTP Headers
3/Toru	Render, PostgreSQL, ORM, JSDoc and Swagger
4/Whā	API Versioning, Content Negotiation, Relationships and Repository Pattern
5/Rima	Validation, Filtering and Sorting
6/Ono	Seeding and API Testing
7/Whitu	Logging, Authentication and Jail
8/Waru	Role-Based Access Control
Mid Semester Break	
9/Whitu	Rate Limiting and Securing HTTP Headers
10/Tekau	ERD, Documentation and Software Methodologies
11/Tekau mā tahi	Project Work and Practical Work Due
12/Tekau mā rua	Project Work
13/Tekau mā toru	Project Work
14/Tekau mā whā	Project Work
15/Tekau mā rima	Project Work Due
16/Tekau mā ono	Catch Up Week

Resources

Software

This paper will be taught using **Microsoft Visual Studio Code** and **Node.js**. An installer for **Microsoft Visual Studio Code** and **Node.js** are available - https://code.visualstudio.com/download and https://nodejs.org/en/download. Please refer any problems with downloads or installers to **Rob Broadley** in D205a.

Readings

No textbook is required for this course. URLs to useful resources will be provided in the lecture notes.

Course Requirements and Expectations

Learning Hours

This course requires **150 hours** of learning. This time includes **60 hours** directed learning hours and **90** self-directed learning hours.

Criteria for Passing

To pass this paper, you must achieve a cumulative pass mark of **50%** over all assessments. There are no reassessments or resits.

Attendance

- Learners are expected to attend all classes, including lectures and labs.
- If you cannot attend for a few days for any reason, contact the course.

Communication

Microsoft Outlook/Teams are the official communication channels for this course. It is your responsibility to regularly check **Microsoft Outlook/Teams** and GitHub for important course material, including changes to class scheduling or assessment details. Not checking will not be accepted as an excuse.

Snow Days/Polytechnic Closure

In the event **Otago Polytechnicmessage** is closed or has a delayed opening because of snow or bad weather, you should not attempt to attend class if it is unsafe to do so. It is possible that the course lecturer will not be able to attend either, so classes will not physically be meeting. However, this does not become a holiday. Rather, the course material will be made available on GitHub for classes affected by the closure. You are responsible for any course material presented in this manner. Information about closure will be posted on the **Otago Polytechnicmessage Facebook** page https://www.facebook.com/OtagoPoly.

Group Work and Originality

Learners in the **Bachelor of Information Technology** programme are expected to hand in original work. Learners are encouraged to discuss assessments with their fellow learners, however, all assessments are to be completed as individual works unless group work is explicitly required (i.e. if it doesn't say it is group work then it is not group work - even if a group consultation was involved). Failure to submit your original work will be treated as plagiarism.

AI Tools

Learning to use AI tool is an important skill. While AI tools are powerful, you **must** be aware of the following:

- If you provide an AI tool with a prompt that is not refined enough, it may generate a not-so-useful response
- Do not trust the AI tool's responses blindly. You **must** still use your judgement and may need to do additional research to determine if the response is correct
- Acknowledge what AI tool you have used. In the assessment's repository **README.md** file, please include what prompt(s) you provided to the AI tool and how you used the response(s) to help you with your work

Referencing

Appropriate referencing is required for all work. Referencing standards will be specified by the course lecturer.

Plagiarism

Plagiarism is submitting someone elses work as your own. Plagiarism offences are taken seriously and an assessment that has been plagiarised may be awarded a zero mark. A definition of plagiarism is in the Student Handbook, available online or at the school office.

Submission Requirements

All assessments are to be submitted by the time, date, and method given when the assessment is issued. Failure to meet all requirements will result in a penalty of up to **10%** per day (including weekends).

Extensions

Familiarise yourself with the assessment due dates. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted for poor time management or pressure of other assessments.

Impairment

In case of sickness contact the course lecturer or **Head of Information Technology (Michael Holtz)** as soon as possible, preferably before the assessment is due. The policy regarding the granting of a mark that considers impaired performance requires a medical certificate and a medical practitioner's signature on a form. You may refer to the guide on impaired performance on the student handbook.

Appeals

If you are concerned about any aspect of your assessment, approach the course lecturer in the first instance. We support an open-door policy and aim to resolve issues promptly. Further support is available from the **Head of Information Technology (Michael Holtz)** and **Second/Third-Year Coordinator (Grayson Orr)**. **Otago Polytechnicmessage** has a formal process for academic appeals if necessary.

Other Documents

Regulatory documents relating to this course can be found on the **Otago Polytechnicmessage** website.